Research on Intelligent Image Segmentation Based on UAV Remote Sensing and DeepLabV3+ Model

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***Abstract*—Semantic segmentation of UAV remote sensing crop images faces the dual challenges of high-accuracy models that are difficult to deploy in edge devices and the impaired accuracy of lightweight networks. In this paper, based on DeepLabV3+, VGG16, ResNet50, ResNet101, Xception, and MobileNetV2 are used as the backbones for comparative experiments, and compared and analyzed in terms of segmentation accuracy (pixel accuracy, IoU, Dice) and computational efficiency (memory occupation, inference speed). The results show that: Xception and MobileNetV2 perform best in terms of performance and efficiency, while MobileNetV2 is more suitable for deployment in resource-constrained scenarios such as unmanned aerial vehicles (UAVs), considering the accuracy, model scale, stability, and generalization ability, which provides a feasible solution for efficient semantic segmentation of crop remote sensing images.**

Keywords- DeepLabV3+, Deep Learning, Semantic Segmentation, Agriculture

I. INTRODUCTION

Agricultural production is the basis for food security and rural stability, and its statistical surveys are of great significance to national development. However, traditional manual surveys are inefficient and data quality is difficult to guarantee. With the development of UAV remote sensing and deep learning, crop distribution extraction based on semantic segmentation has become an important tool for intelligent agricultural decision-making [1].

At present, the research on image semantic segmentation technology based on deep neural networks has made breakthrough progress, The DeepLabV3+ model with its excellent performance and wide adaptability, has been widely used in smart agriculture. Existing studies include: Chen L-C et al. proposed DeepLabV3+ [2], fused with Xception decoder to optimize multi-scale feature interaction; Fu Bihuan et al. used ResNet et al. for semantic segmentation of predicted images, and obtained mIoUs higher than 90% [3]; and Mu Taoyang [4] et al. combined with remote sensing of unmanned aerial vehicles to achieve rice fall recognition (IoU 93%); The VGG network structure had been proposed as an example by Long [5], laying the foundation of semantic segmentation research based on convolutional networks.

This paper uses DeepLabV3+ with five different backbone networks to compare segmentation performance and select the best one.

II. RESEARCH METHODS AND DATA HANDLING

A. Neural Network Research Methods

This paper conducts a controlled-variable comparison of five backbones—VGG16, ResNet50, ResNet101, Xception, and MobileNetV2—within the DeepLabV3+ segmentation framework. To ensure methodological consistency under the principle of a single variable, a uniform configuration was adopted: All experiments used Adam (lr=0.001), cross-entropy loss, batch size 4, 50 epochs; dataset split 70%/30%.

B. Data acquisition and pre-processing

1) Data Acquisition and the Potential Impact of the Characteristics of the Data.

Data are sourced from AliCloud Tianchi[6]: GB-scale UAV images of farmland in Xingren, Guizhou, annotated at the pixel level for five classes (barley, maize, tobacco, Job's tears, buildings). Since the data were derived from the AliCloud platform, they lacked precise weather, light, shadow distribution, and growth information, and these uncertainties may lead to a decrease in the classification accuracy of the model and errors in texture feature extraction. This study did not compensate for this at present. In the future, the robustness of the model in complex environments and throughout the full fertility cycle should be evaluated and improved by supplementing the data with diverse meteorological (e.g., cloudy, rainy, hazy) conditions, various time periods (early morning, and nighttime), and cross-seasonal aerial photographs.

2) Data Preprocessing.

Given the huge volume of the original large image and the high processing cost, Original images and labels were tiled into 1024×1024 patches; patches with only background pixels were discarded.

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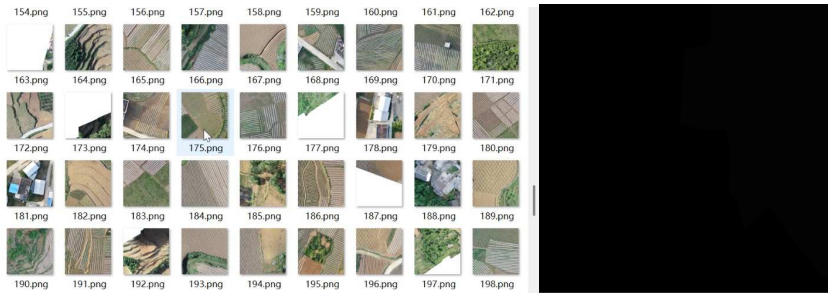


Figure 1. Data after Preprocessing (not fully shown) and Sample Mask Image(foreground=#040404)

C. *Network Hyperparameters*

1) Loss Function and Optimizer.

The Adam optimizer and cross-entropy loss function.

2) Class Weights and Image Pooling.

To mitigate the problem of class imbalance in the dataset, different weights are set for the background and foreground when computing the loss: class_weight = torch.tensor([1.0, 2.0, 2.0, 2.0, 2.0, 2.0]). To reduce the training overhead, the image is scaled down from 1024*1024 to 224*224. For the original image, bilinear interpolation is used, whereas for the annotated image, nearest neighbor interpolation is taken to keep the mask colors constant.

III. NETWORK ARCHITECTURE

A. *Overall Model Architecture*

A standard semantic segmentation framework typically adopts an encoder-decoder architecture. The encoder usually employs a pre-trained backbone network (e.g., VGG, ResNet) for feature extraction, followed by a decoder network for classification, in this paper DeepLabV3+.

B. *DeepLabV3+ Model*

DeepLabV3+ uses dilated (atrous) convolutions and an Atrous Spatial Pyramid Pooling (ASPP) module to capture multi-scale context. ASPP contains parallel branches of dilated convolutions with different dilation rates and a global pooling branch, whose outputs are concatenated. The formula is shown as (1).

$$\boldsymbol{y}[\boldsymbol{i}] = \sum_{\boldsymbol{k}} \boldsymbol{x}[\boldsymbol{i} + \boldsymbol{r} \cdot \boldsymbol{k}]\boldsymbol{w}[\boldsymbol{k}] \tag{1}$$

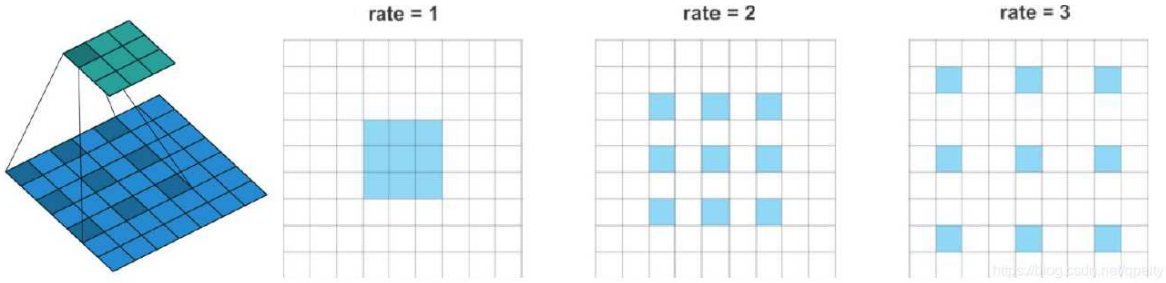


Figure 2. Schematic Diagram of the Atrous Convolution

Atrous Spatial Pyramid Pooling (ASPP) is designed as a multi-scale feature extraction module in DeepLabV3+. ASPP consists of four parallel branches [2]: three Atrous Convolutions with three expansions (r = 6, 12, 18) and a global average pooling branch. The global pooling first downscales the input by 1×1 convolution, then up-samples the result to the original feature map size by bilinear interpolation, and the final output concated with the output of each Atrous Convolution.

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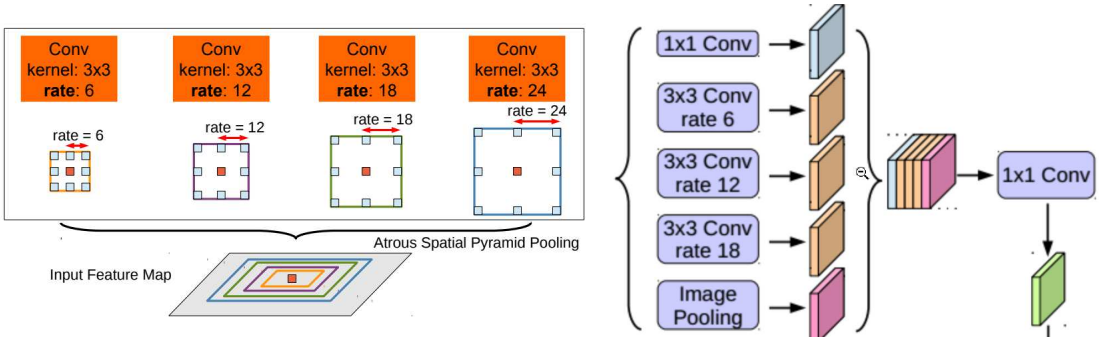


Figure 3. ASPP-Arbeitsprinzip-Diagram

The DeepLabV3+ architecture is shown in Figure 4. Its decoder is connected in series with the encoder, while the ASPP module is positioned at the end of the encoder.

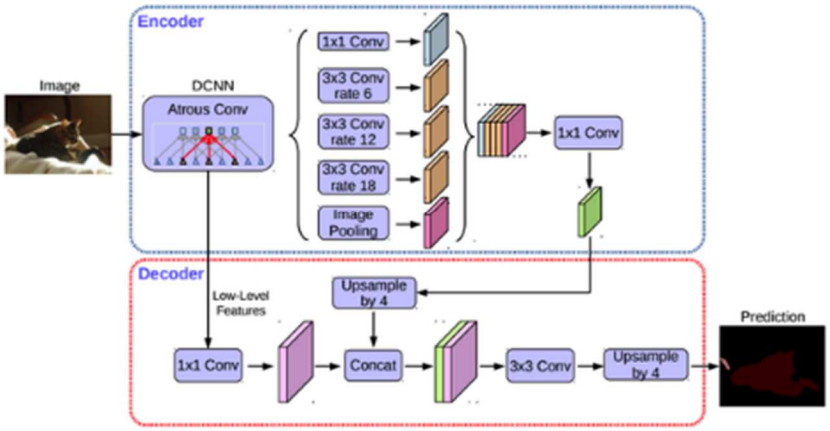


Figure 4. DeepLabV3+ Model Framework and Execution Process

C. *Image Semantic Segmentation Model Architecture Diagram*

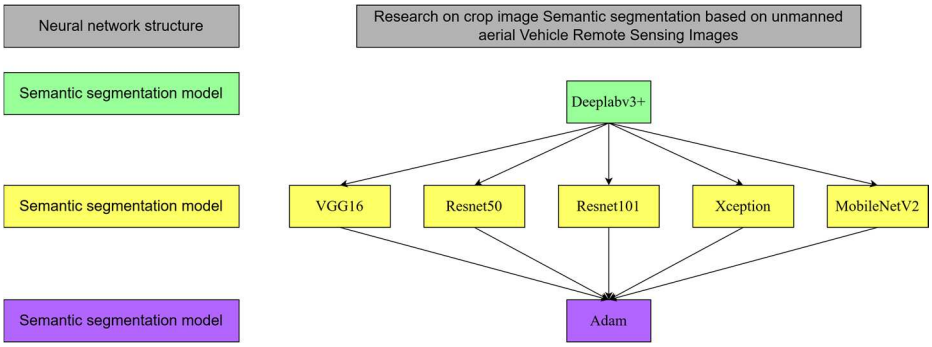


Figure 5. Image Semantic Segmentation Network Architecture

IV. Experiment and Results Analysis

A. *Evaluation Metrics*

This experiment evaluates models by both cost (learning rate, GPU memory usage, and inference time) and performance (loss, accuracy, IoU, Dice) metrics. To evaluate models' generalization ability, the Normative Generalization Gap (NGG) is defined as:

$$NormalizedGeneraliaztionGap = \frac{TrainAcc - ValAcc}{TrainAcc} \quad (2)$$

Due to the relatively large variance of much of the data, sliding window means were used frequently to smooth out the curves. To assess the stability of the model performance, a regression was conducted on the smoothed curves of performance over rounds, and then the residual sum of squares (RSS) was calculated as an indicator of stability.

B. *Experimental results and analysis*

1) Dynamic Accuracy Visualization and Analysis.

The first metric that needs to be evaluated is the correctness rate, and for this purpose, curves of accuracy and Dice coefficients on the training set across epochs were plotted for the models (we analyze the metrics on the training set firstly because it is much more stable and hence

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is suitable for initial judgments).

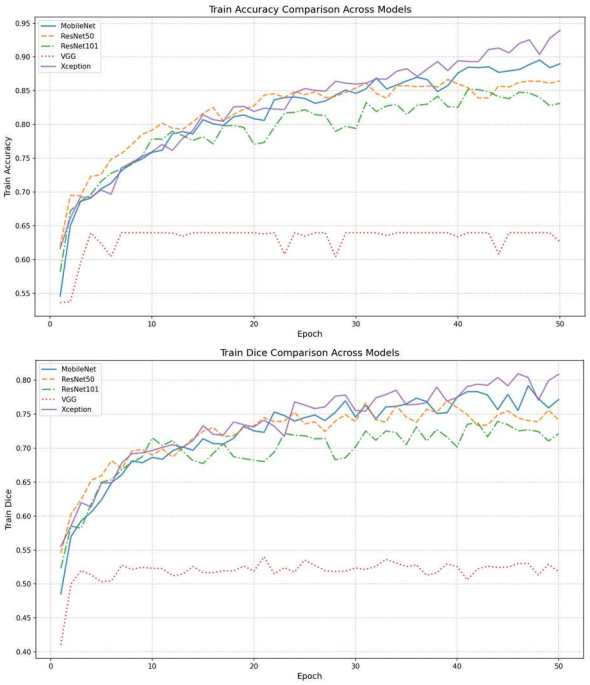


Figure 6. Curves of Train Accuracy and Train Dice across Epochs

Training accuracy and Dice curves (Fig. 6) showed VGG plateauing early due to its overly simplistic architecture; the other models (except Xception and MobileNetV2) leveled off around epoch 40. Xception continued to increase (near 95% accuracy at the end), suggesting potential overfitting; MobileNetV2 also increased steadily but more slowly. To evaluate each model's actual performance more objectively, the following section presents a comparative analysis of dynamic accuracy on the validation dataset across training epochs.

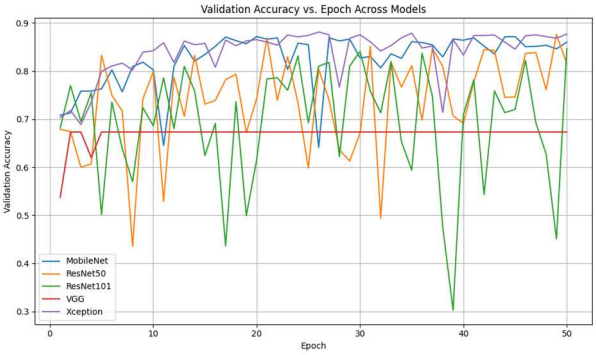


Figure 7. Curves of Accuracy across Epochs

Initial analysis shows that Xception and MobileNetV2 are close in val acc at the end of the training, confirming that Xception may have significant overfitting compared to the latter. In order to visualize the degree of overfitting of the different models the following graphs were painted:

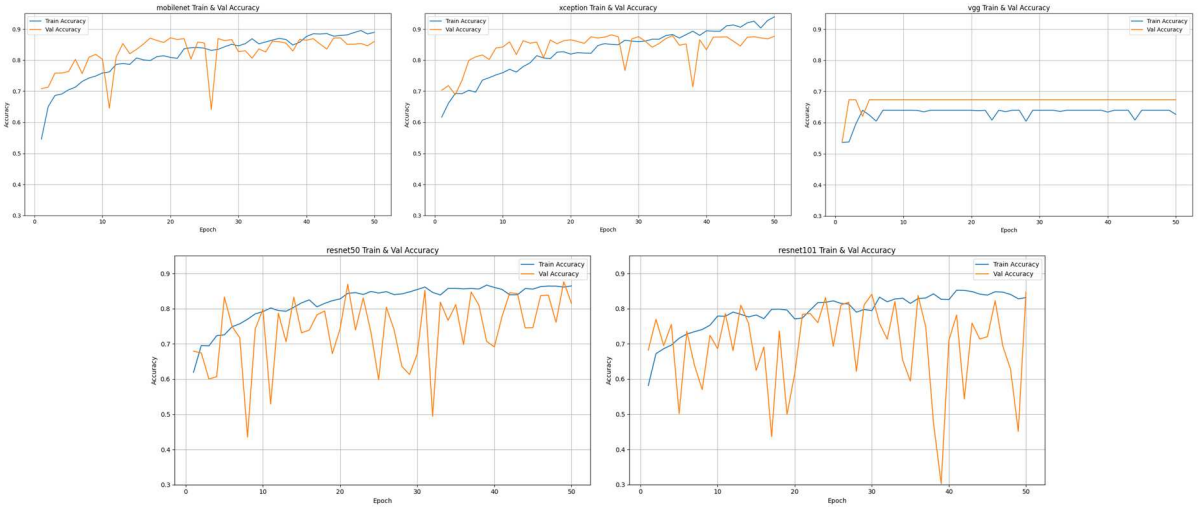


Figure 8. Accuracy Curves on Training and Validation Datasets

During the early training phase, MobileNetV2 and Xception both had markedly higher validation accuracy than training, likely because Dropout and BatchNorm were active during training—suppressing performance—whereas during validation Dropout was disabled and BatchNorm utilized global statistics, allowing the models to perform at full capacity. In the late training phase (epoch ≥ 30), the difference (val acc − train acc) becomes negative and declines monotonically, indicating the onset of overfitting. In contrast, the ResNet variants exhibit large swings in validation performance and poor stability. To quantify each model's generalization, let's measure the NGG, whose larger values signify a weaker ability of generalization and more severe overfitting

As shown in Figure 9, the ResNet variants exhibit markedly poorer generalization, whereas MobileNetV2 and Xception deliver similarly high and stable generalization ability. This contrast stems from ResNet's excessive parameter count, which encourages memorization of irrelevant details; by contrast, the more compact MobileNetV2 and Xception better matched the complexity of this task, confirming that model capacity should align with the problem difficulty. Furthermore, in the final ten epochs, Xception exhibits a slightly higher level of overfitting than MobileNetV2, which aligns with our previous hypothesis. Notably, despite Xception owning roughly five times as many parameters as MobileNetV2, the degradation in its generalization

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performance is disproportionate, which can be explained by its Depthwise Separable Convolutions' parameter sharing, and frequent usage of BatchNorm layers that impose stronger regularization[7].

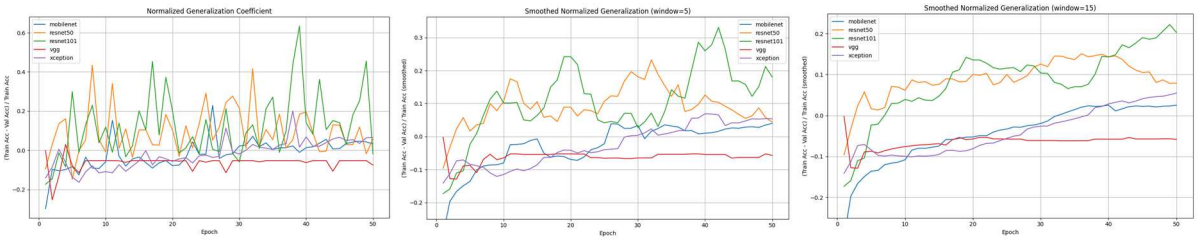


Figure 9. Curves of NGG across Epochs and Curves of the Sliding Window Means of NGG

2) Visual Analyses of Performance Stability.

Next, the learning curves (smoothed val acc versus epoch) of different models are going to be compared. The sliding window mean curve of val acc:

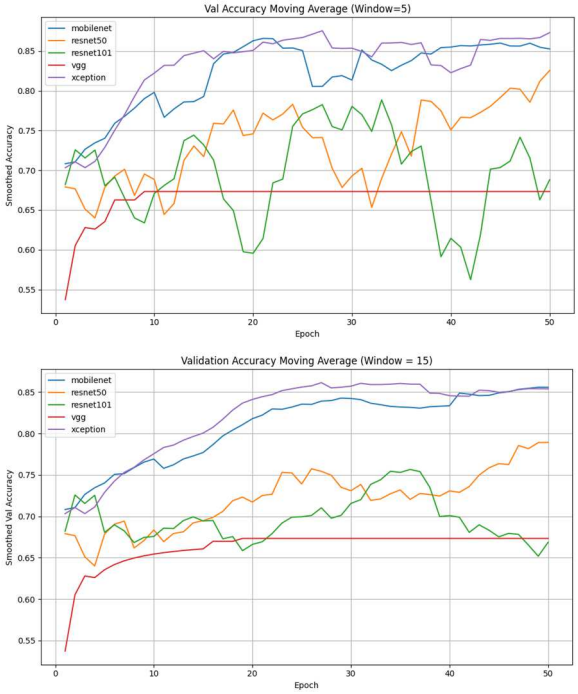


Figure 10. Curves of Sliding Window Mean of Val Acc when window_size=5, 15

During early training, Xception, MobileNetV2, and VGG all show rapid gains; thereafter, Xception and MobileNetV2 slow and stabilize, while VGG prematurely stalls into a local optimum. By contrast, the ResNet family's curves are almost linear—ResNet50 approximates a straight line, and ResNet101 hovers around 70% (with large fluctuations, though). This suggests that increasing parameter count leads to slower learning speed and reduced stability. We quantified model stability by fitting these curves (logarithmic fits for Xception, MobileNetV2, VGG; linear fits for ResNet50/101) and analyzing residuals to give information about the rationality of our choice of models.

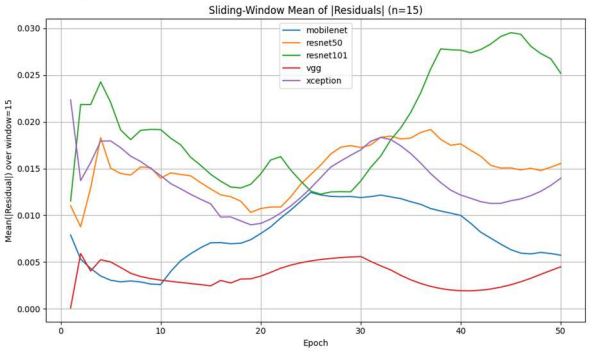


Figure 11. Sliding-Window Mean of |Residuals| (window_size=15)

Table 1. Comparison of Model Fit Statistics and Parameter Count

	VGG	MobilenetV2	Xception	ResNet50	ResNet101
RSS	0.001153[1]	0.004749[2]↑	0.011367[3]	0.015597[4]	0.034442[5]↓
Correlation Coefficients	0.9786	0.9732	0.9507	0.8697	0.0750
Quantity of Parameters	140M	4.25M	22.91M	25.64M	44.71M

Due to VGG16's extremely poor performance, it is excluded from the primary comparison. The abnormal correlation coefficient of ResNet101 can be considered a consequence of its extreme instability because that of ResNet50 also slightly underperforms the lightweight models. Excluding VGG16, model stability generally increases with the parameter count. VGG16's anomalous behavior likely stems from its highly symmetrical, non-residual architecture (which hinders gradient flow in deep layers) and overly aggressive regularization of dropout=0.5. Overall, model stability ranks as follows: MobileNetV2 > Xception > ResNet50 > ResNet101. Larger models tend to overfit due to parameter redundancy, leading to pronounced fluctuations in validation performance. The easiest way to mitigate this problem is through data augmentation.

3) Quantitative Analysis of Overall Correctness.

In the final-epoch evaluation, the ideal approach is to

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compute the mean and standard deviation of each metric from multiple independent runs; however, due to limited computing resources and time, this study instead estimates these statistics using the data from the last ten epochs. For clarity, standard deviations ≤ 0.002 are highlighted in green, ≥ 0.003 in red, and values in between in light red.

Table 2. Comparison of Training and Validation Performance and Standard Deviation of Each Model

Model	Train Acc	Std Dev	Train IoU	Std Dev	Train Dice	Std Dev	Val Acc	Std Dev	Val IoU	Std Dev	Val Dice	Std Dev
MobileNetV2	0.8851[2]	0.0053	0.7274[2]	0.0126	0.7732[2]	0.0124	0.8562[2]	0.0118	0.7019[1]↑	0.0274	0.7499[1]↑	0.0246
ResNet50	0.8562[3]	0.0096	0.6964[3]	0.0088	0.7443[3]	0.0080	0.8085[3]	0.0471	0.6405[3]	0.0499	0.6933[3]	0.0487
ResNet101	0.8426[4]↓	0.0083	0.6777[4]↓	0.0104	0.7272[4]	0.0095	0.6958[4]↓	0.1243	0.5734[4]↓	0.0977	0.6302[4]↓	0.0964
VGG	0.6352[5]	0.0104	0.4974[5]	0.0076	0.5224[5]	0.0078	0.6733[5]	0.0000	0.5275[5]	0.0000	0.5502[5]	0.0000
Xception	0.9133[1]↑	0.0150	0.7550[1]↑	0.0129	0.7965[1]	0.0115	0.8695[1]↑	0.0097	0.6998[2]	0.0276	0.7475[2]	0.0260

Although validation metrics are more practically relevant, their reliability was limited by large standard deviations. On the validation set, Xception achieved slightly higher accuracy than MobileNetV2 but slightly lower IoU and Dice; Since the difference in means of performance between the two models is smaller than their standard deviations, the comparative performance can be considered statistically indistinguishable based on the current data. Considering that MobileNetV2 exhibits milder overfitting and lower variance in later epochs, it holds a marginal overall advantage; however, their relative ranking in generalization ability briefly changed twice during training, indicating that conclusions based solely on the final few epochs may lack robustness.

ResNet variants consistently lag behind Xception and MobileNetV2 – both architected with segmentation-specific components like multi-scale feature fusion layers – in performance metrics across training and validation datasets, with ResNet101 persistently underperforming ResNet50. Larger models exhibit steeper learning curves and have a poorer ability of generalization in addition, resulting in inferior performance on relatively simple crop-segmentation tasks compared to lightweight architectures.

4) Quantitative Analysis of the Costs of Different Models.

Table 3. Comparison Table of Model Resource Consumption and Computational Efficiency

	MobileNetV2	ResNet50	ResNet101	VGG	Xception
Memory Usage(MB)	88.4700[1]↓	622.4500[4]	920.5700[5]↓	304.7798[2]	583.7100[3]
Std Dev	0.0000	0.0000	0.0000	0.0014	0.0000
Inference Time(s)	0.0204[4]	0.0167[3]	0.0256[5]↓	0.0040[1]	0.0133[2]↑
Std Dev	0.0048	0.0046	0.0023	0.0011	0.0033
Training Rate	215.8770[4]	263.1002[3]	160.2190[5]↓	1069.3790[1]	325.3308[2]↑
Std Dev	52.1427	56.5837	8.4797	196.9096	52.2157
Quantity of participants	4.25M[1]	25.64M[3]	44.71M[4]	140M[5]	22.91M[2]

For resource usage, VGG16 delivered the fastest training/inference speed but insufficient accuracy; MobileNetV2 had the smallest memory footprint and robust performance, suitable for deployment; ResNet101 used nearly twice the resources of ResNet50 for little gain. MobileNetV2 also controlled overfitting best, while Xception trained and inferred faster; MobileNetV2's depthwise convolutions incur high memory-access overhead, reducing its computational efficiency. Consequently, shallower architectures may be preferable for tasks involving relatively simple datasets or strict resource/real-time constraints.

Moreover, while MobileNetV2 demonstrated superior overfitting control and stability, Xception provided faster training and inference. This difference seemed primarily from MobileNetV2's depthwise separable convolutions, which incurred high memory-access overhead due to frequent intermediate feature-map read/write operations and underutilization of parallelism[8], thereby reducing its overall computational efficiency on GPU platforms.

C. Conclusions

VGG16 is almost unsuitable for these tasks; the ResNet networks had low accuracy and heavy overfitting; MobileNetV2 and Xception were comparable in accuracy, stability, and generalization, with MobileNetV2 having the smallest memory footprint (though about 1.4× the parallel training/inference overhead of Xception). Considering UAV resource constraints, MobileNetV2 is slightly preferred.

V. Summary and Outlook

In this study, based on the DeepLabV3+ framework, we compared five backbone networks for UAV crop image segmentation. MobileNetV2 had an extremely low memory footprint and stable training/validation performance, with accuracy (~86.95%±0.97%) comparable to Xception, significantly reducing

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computational cost and thus being better suited for mobile deployment.

However, this study several has limitations: it did not use data augmentation, causing considerable performance fluctuations for some models on the validation set; and the number of experimental repeats was insufficient (fewer than five), undermining the statistical reliability of our conclusions.

Future work: (1) use statistical and information-theoretic metrics (e.g., distance correlation [9], and maximal information coefficient [10]) to characterize dependencies among performance metrics; (2) apply LOESS [11], generalized additive models (GAM) [12], and elastic net regression [13] to quantify nonlinear relationships between network depth and model performance or cost (based on ResNet variants). This framework aims to establish empirically validated principles for backbone selection focused on accuracy-efficiency trade-offs.

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REFERENCES

[1] Cao Y et al 2024 Crop Disease Monitoring Using UAV Remote Sensing J. Shenyang Agric. Univ.

[2] Chen L C et al 2018 Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation Proc. ECCV arXiv:1802.02611.

[3] Fu B H and Huang L 2022 Extraction of Tobacco Cultivation Area from UAV Images Based on Deep Semantic Segmentation J. Kunming Univ. Sci. Technol. (Nat. Sci. Ed.) 47(2) 7-15

[4] Mu T Y, Zhao W, Hu X Y et al 2022 Rice Lodging Identification via Improved DeepLabV3+ and UAV Remote Sensing J. Northeast For. Univ.

[5] Long J, et al. Fully Convolutional Networks for Semantic Segmentation [C].DOI: 10.1109/CVPR.2015.7298965.

[6] Tianchi, "Barley Remote Sensing Dataset" [Online Dataset]. 2020. Available: https://tianchi.aliyun.com/dataset/dataDetail?dataId=74952

[7] Lange S, Helfrich K, Ye Q 2022 Batch Normalization Preconditioning for Neural Network Training. J. Mach. Learn. Res. 23:1–43.

[8] Sandler M et al. 2018 MobileNetV2: Inverted Residuals and Linear Bottlenecks. Proc. CVPR 4510–4520

[9] Székely G J, Rizzo ML, Bakirov N K 2007 Measuring and Testing Dependence by Correlation of Distances. Ann. Stat.

[10] Reshef D N et al. 2011 Detecting Novel Associations in Large Data Sets. Science 334(6062)

[11] Cleveland W S 1979 Robust Locally Weighted Regression and Smoothing Scatterplots. J. Am. Stat. Assoc.

[12] Hastie T, Tibshirani R 1990 Generalized Additive Models

[13] Zou H, Hastie T 2005 Regularization and Variable Selection via the Elastic Net. J. R. Stat. Soc.

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